WNBA Player Projection Analysis

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Date: March 11, 2026

Agenda

- Project Introduction
- Data Processing and Analysis
- Key Evaluation Metrics
- Key Findings & Insights
- Power BI Dashboard Highlights
- Recommendations
- Q&A

Project Introduction

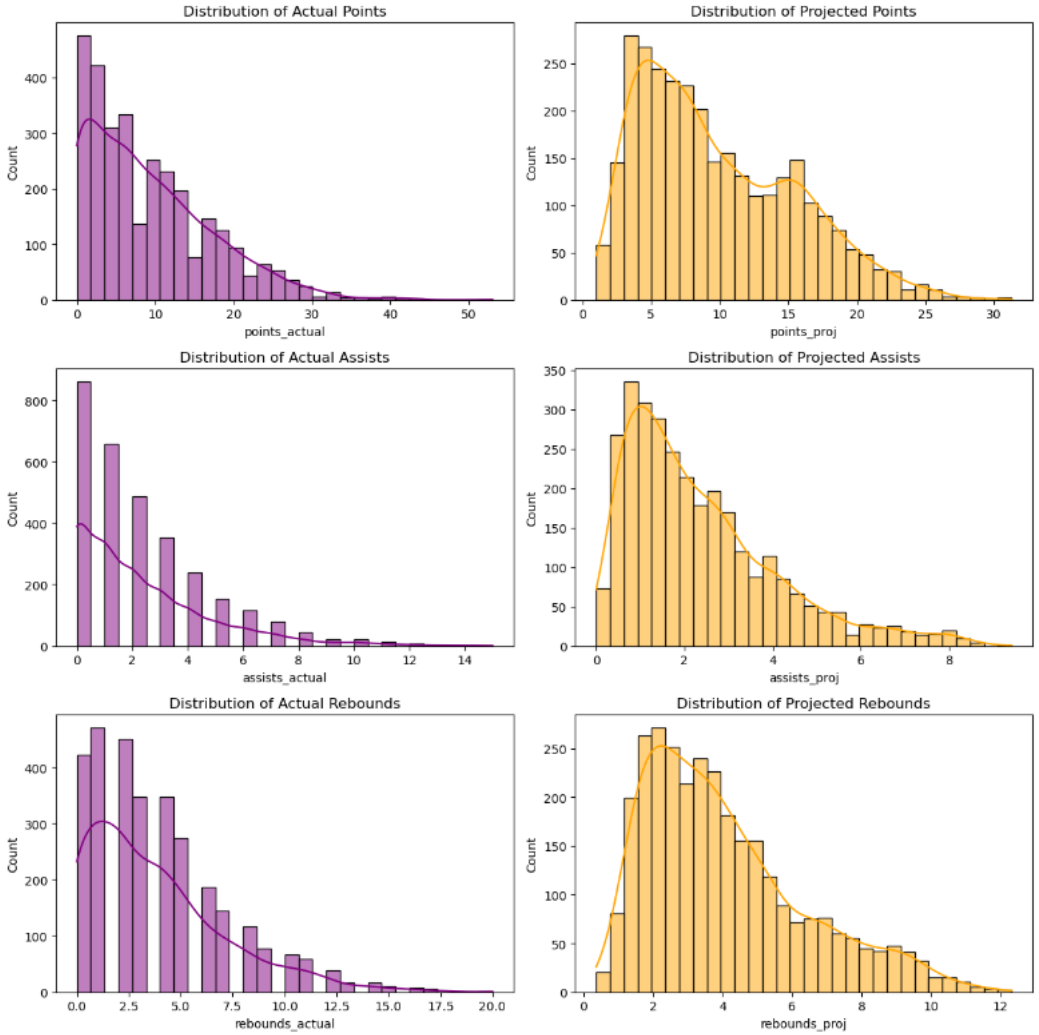
- **Objective**: Evaluate the accuracy of WNBA player projections for the 2023 season.
- **Scope**:
  - Compare actual vs. projected player performance (points, assists, rebounds).
  - Assess model accuracy and identify areas for improvement.
  - Provide actionable recommendations for AI/ML and business teams.
- **Tools Used**: Python (Jupyter Notebook), Power BI

Data Processing and Analysis

- **Three datasets used:**
  - Player projections (2023_WNBA_player_projections.csv)
  - Team-level box scores (2023_WNBA_team_box_score.csv)
  - Player-level box scores (2023_WNBA_player_box_score.csv)
- **Processing steps:**
  - Merging and Cleaning: Merged three datasets, included only played players and relevant columns.
  - Feature Engineering: Computed projection errors for points, assists and rebounds.

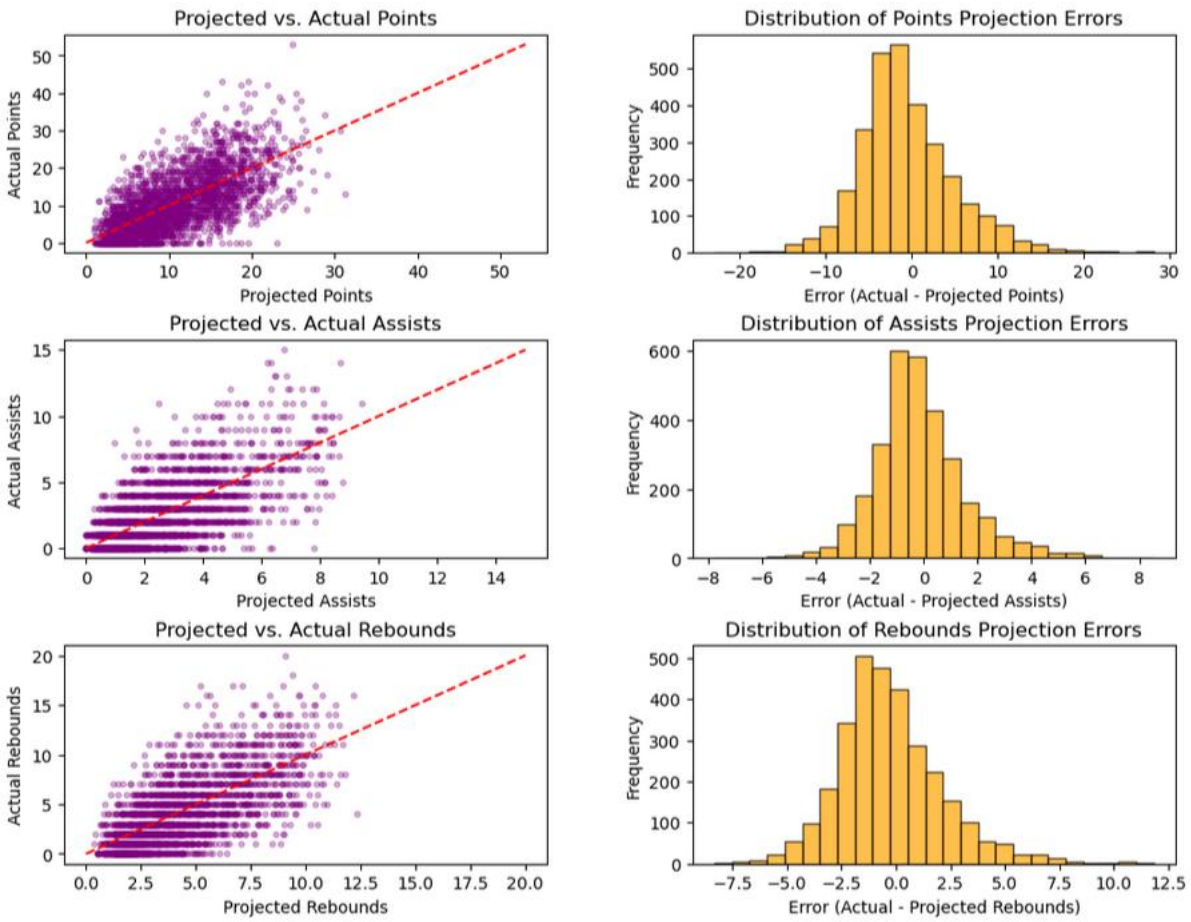
Data Processing and Analysis – Continue

- **Distribution analysis (actual vs. projections)**
  - Actual data exhibited heavy-tailed right skewed distributions.
  - Projected data exhibited more concentrated distributions; the model appeared to predict fewer extreme values.



Data Processing and Analysis – Continue

- **Distribution analysis of projection errors**
  - Points, assists and rebounds errors exhibited approximately bell-shaped distributions but with heavier tails than a normal distribution, indicating a higher frequency of extreme errors.



Key Evaluation Metrics

- **Mean Absolute Error (MAE) – Primary Metric**
  - **Why?** Measures the absolute error. MAE is robust to non-normal, heavy-tailed distributions. Stable and interpretable for skewed data and extreme values.
- **Median Absolute Percentage Error (MDAPE) – Secondary Metric**
  - **Why?** Measures the percentage error. Use the median instead of the mean, making it more robust than MAPE and less sensitive to small values.
- **Weighted Absolute Percentage Error (WAPE) – Secondary Metric**
  - **Why?** Measures the percentage error. Balance the impact of errors across high and low-performing players.

Key Findings & Insights

- **Overall Projection Accuracy**

Metrics	Points	Assists	Rebounds
MAE	4.4	1.3	1.8
MDAPE	36.6%	40.7%	37.3%
WAPE	47.4%	58.8%	48.9%

  - **MAE values are within industry benchmarks (points 3-5, assists 1-2 and rebounds 1.5-2), indicating acceptable absolute errors.**
  - **High MDAPE and WAPE reflect significant error variability, suggesting room for improvement.**

Key Findings & Insights – Continue

- **Projection Accuracy by Event Type**

Metrics	Points		Assists		Rebounds	
Event Type	Regular	Playoff	Regular	Playoff	Regular	Playoff
MAE	4.3	4.8	1.3	1.3	1.8	2.0
MDAPE	36.5%	38.1%	41.2%	35.8%	37.2%	37.5%
WAPE	47.5%	47.4%	59.3%	54.9%	49.7%	43.8%

- **Event type (regular season vs. playoffs) has a minimal impact on projection accuracy.**

Key Findings & Insights – Continue

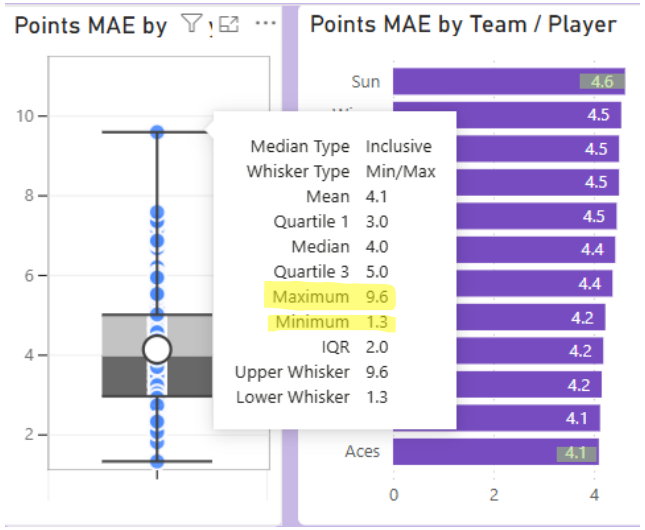
- **Projection Accuracy by Starter vs. Bench Player**

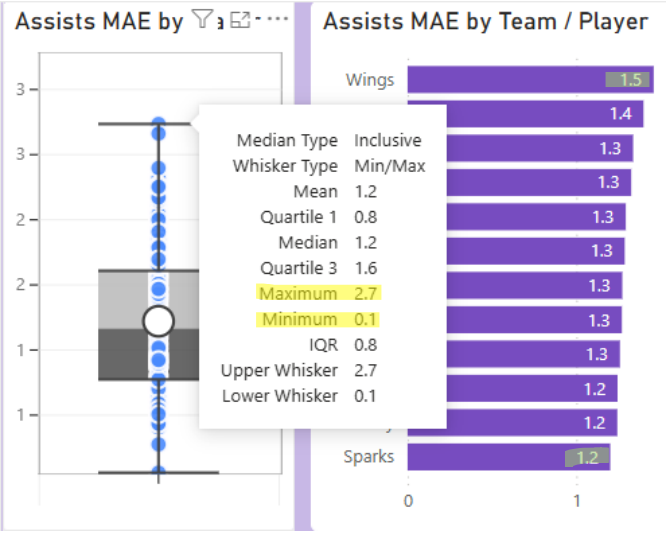
Metrics	Points		Assists		Rebounds	
Starter Status	Starter	**Bench**	Starter	**Bench**	Starter	**Bench**
MAE	5.1	**3.4**	1.6	**1.0**	2.1	**1.5**
MDAPE	32.4%	**46.1%**	38.7%	**43.5%**	33.5%	**44.2%**
WAPE	38.9%	**80.7%**	48.8%	**100.7%**	40.8%	**75.0%**

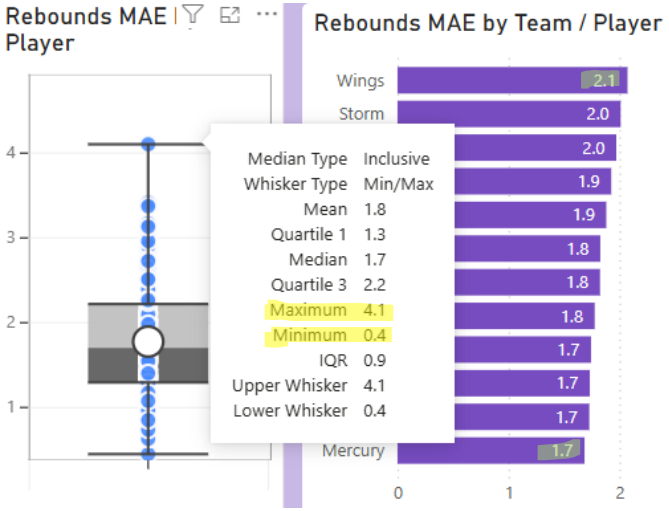
- **Starter status has a major impact on projection accuracy. Bench players have lower absolute errors but much higher relative errors than starters.**
- **The model is far more reliable for starters. Bench players are harder to predict due to fluctuating playing time.**

Key Findings & Insights – Continue

- **Team-Level vs. Player-Level Accuracy**

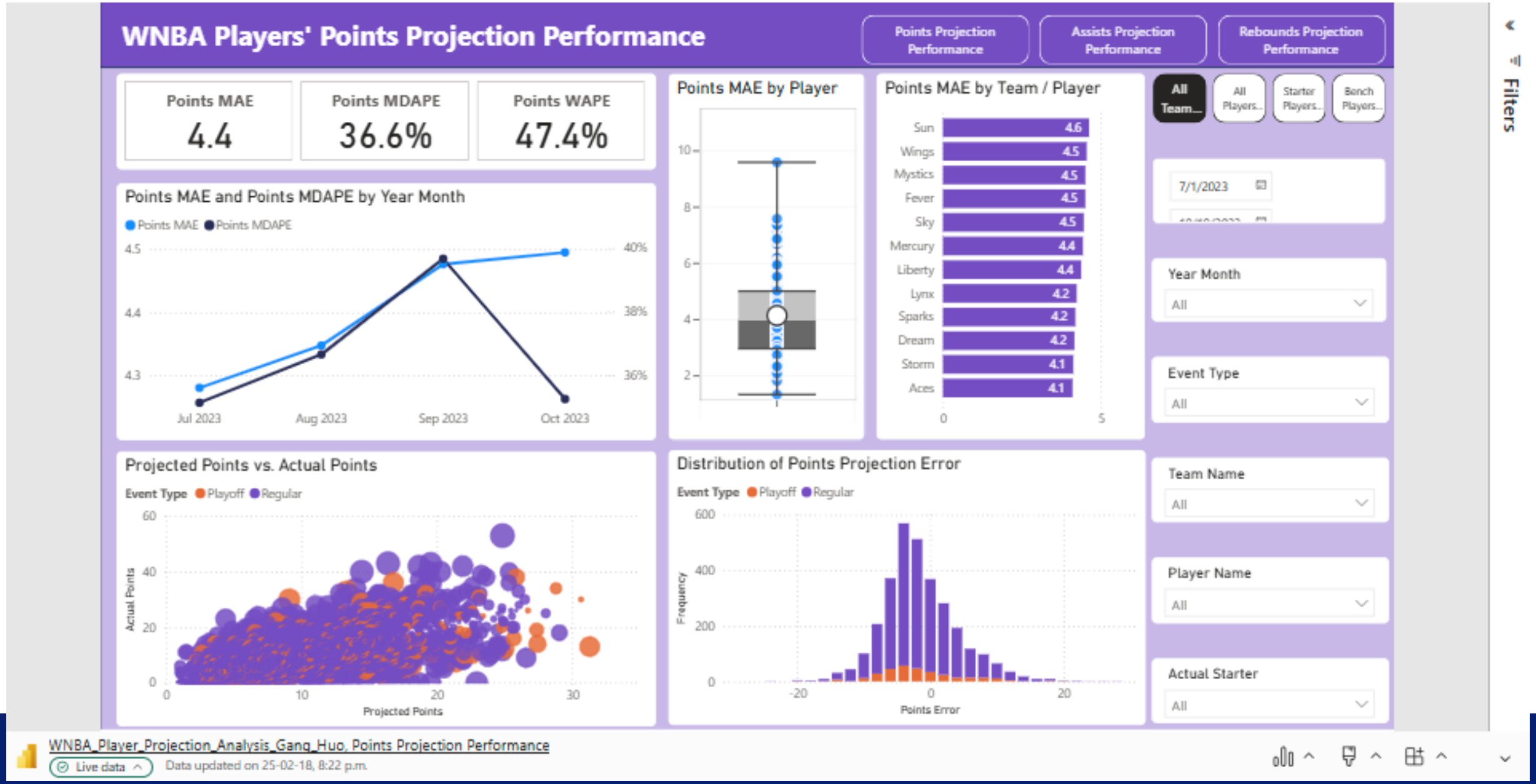






- **MAE values are consistent across teams but vary significantly between individual players. Some players have much higher projection errors. This suggest that high-variance players require more tailored modeling approaches (e.g., position-based) to improve accuracy.**

Power BI Dashboard Highlights



Recommendations

- Utilize ensemble methods to minimize extreme prediction errors and improve prediction consistency.
- If FPSM provides feature importance scores, analyze which features contribute most to projection errors and adjust accordingly.
- If FPSM currently uses a single global model, transition to individualized or role-specific models, with separate models for starters and bench players.

Q&A

Thank you!